Checkpoint C1

Monday, October 26, 2026 4:48 PM

a) For Protocol A

G has some zeros and ones, added to I changes the main diagonal from all 0's to all 1's.

$$x_i[k+1] = \frac{(I+G)_i}{1+d_i} x[k]$$

filters only robots with connections

$1+d_i$ = total # of connections, including itself

b) $\frac{(I+G)_i}{1+d_i} \cdot 1 = 1$

since every value in every row = 1, numerator doesn't filter anything out, results in length of the row

also results in length, since every robot is connected to every robot

ends up being $\frac{1}{1} \cdot 1$

c)

$$(I+D)(1) = \begin{bmatrix} 1+d_1 & 0 & 0 \\ 0 & 1+d_2 & 0 \\ 0 & 0 & 1+d_3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+d_1 \\ 1+d_2 \\ 1+d_3 \end{bmatrix}$$

each entry = # of connections for that robot, including itself

$$[1+d_1] \cdot \frac{(I+G)_i}{1+d_1} = 1+d_1$$

ends up being the # of connections of robot 1 (d_1) + itself (1)

$$\cancel{1+d_1} \cdot \frac{1+d_1}{\cancel{1+d_1}} = 1+d_1$$

of connections each robot has will not change as k ↑, since $(I+D)(1)$ does not change

Protocol B

For ϵL, ϵ divides each non zero term by the max and represents the size of each step

$$x_i[k+1] = (1-\epsilon d_i)(x_i[k]) + \epsilon \sum x_j[k]$$

whichever entries in x are considered neighbors

How much robot "i" will stay in place, if few neighbors then $1-\epsilon d_i$ close to 1

weighted average of neighbors positions, few connections means this value is low.

b)

becomes d_i since every entry = 1

$x_j[k]$ represents all the robot's positions that are neighbors

$$(1-\epsilon d_i)(1) + (\epsilon \sum x_j[k])(1) = 1$$

$$(1-\epsilon d_i)(1) + (\epsilon d_i)(1) = 1$$

$$1 - \frac{d_i}{m} + \frac{d_i}{m} = 1$$

$$1 = 1$$

If the average position is where all the robots are, then the robots won't move.

c)

$$1^T(I-\epsilon L) = 1^T$$

$$1^T - 1^T \epsilon L = 1^T$$

each row

$$1^T - \epsilon(1^T D - 1^T G) = 1^T$$

$$1^T - \epsilon(d - d) = 1^T$$

row vector

since $G = G^T$, sum of columns = sum of rows = d

$$1^T - 0 = 1^T$$

W_B stays constant, therefore the weighted averages for each robot and its neighbor's positions stay the same

d) With $\epsilon > \frac{1}{\max d_i}$, the rows wouldn't sum to 1. Additionally, for the robots with connections close to max d_i, there is a chance that their "stay in place" weight could be negative. Hence, the robot would keep oscillating and never stay in one spot.